Title: AI Student Assistant – An Intelligent Learning Support System

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Introduction

The rapid growth of Artificial Intelligence has transformed education by providing intelligent learning tools. This project develops an AI Student Assistant that integrates machine learning-based NLP techniques with a full-stack web application to assist students in academic learning.

The system provides interactive tools for question answering and writing evaluation, making learning more efficient and personalized.

Objectives

- To build an AI-based student assistant system
- To develop a full-stack web application
- To integrate NLP for question answering and writing evaluation
- To implement secure authentication and user management
- To store user interaction history using a database

System Design

Frontend:

A React-based interface allows users to:

- Register and login
- Ask academic questions
- Submit writing tasks
- View AI responses

Backend:

FastAPI handles:

- API requests
- Authentication (JWT)
- AI model communication
- Database operations

Database:

MongoDB stores:

- User credentials
- Question history
- Writing evaluation results

AI Methodology

The system uses NLP-based AI techniques:

Question Answering:

User input is processed and sent to an AI model which returns contextual answers.

Writing Evaluation:

The AI model analyzes:

- Grammar
- Structure
- Vocabulary
- Coherence

It returns:

- Score (1–9 scale)
- Feedback
- Corrections

Implementation Details

Authentication:

- Users register with username and password
- Passwords are hashed using bcrypt
- JWT tokens are used for secure login sessions

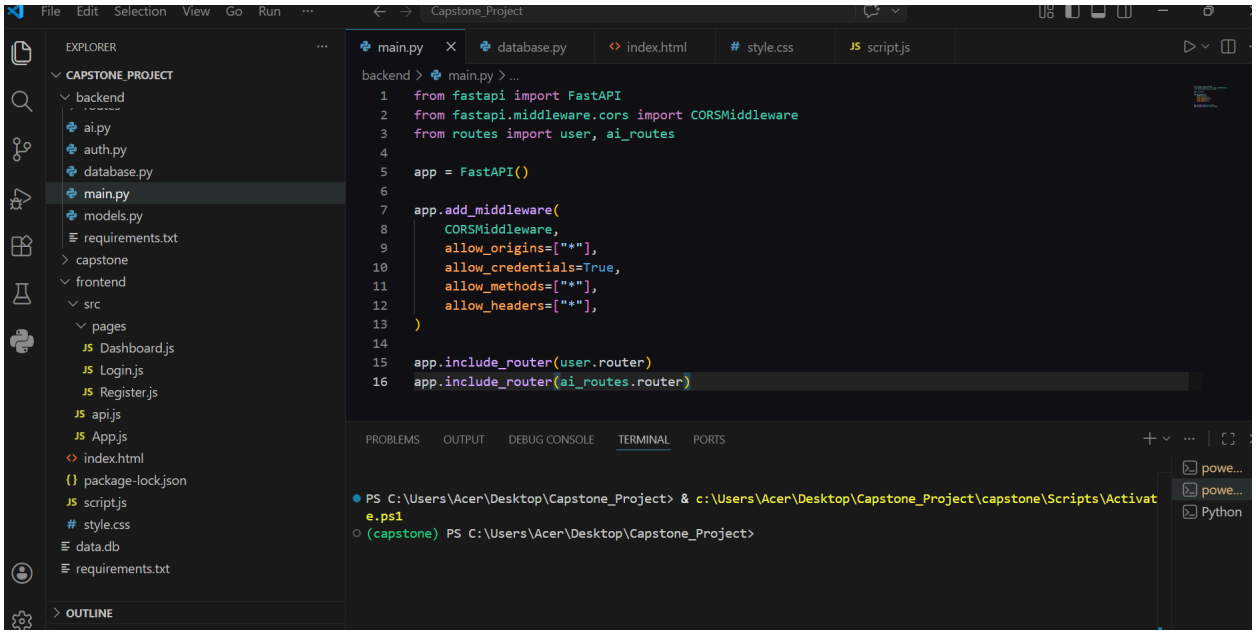
Backend APIs:

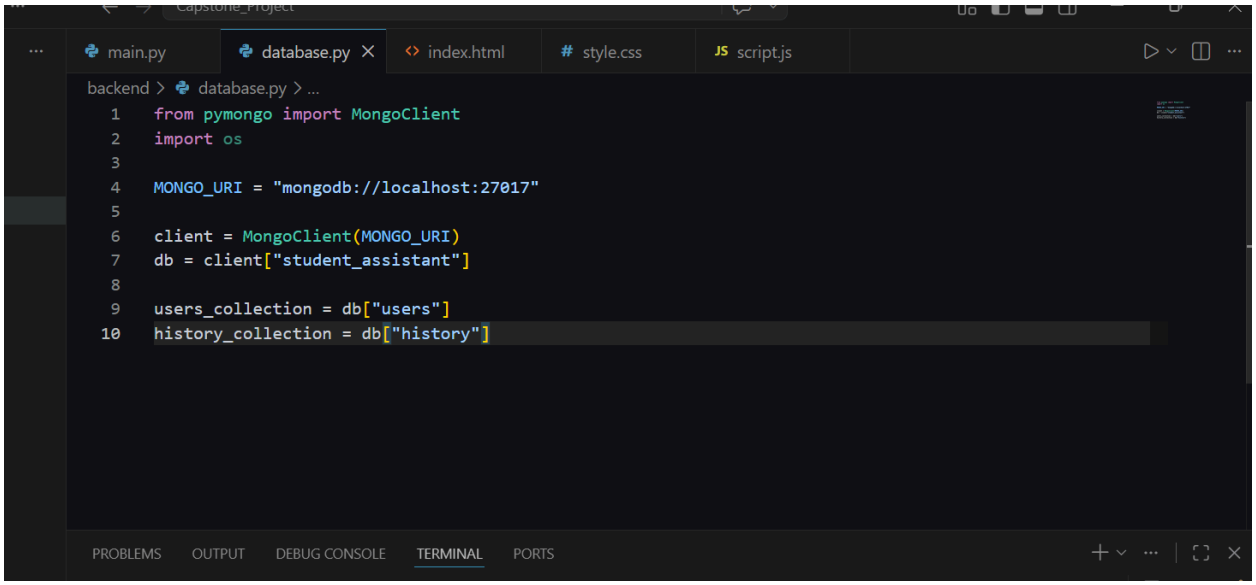
- `/register` → User registration
- `/login` → User authentication
- `/ask` → AI question answering
- `/writing` → Writing evaluation

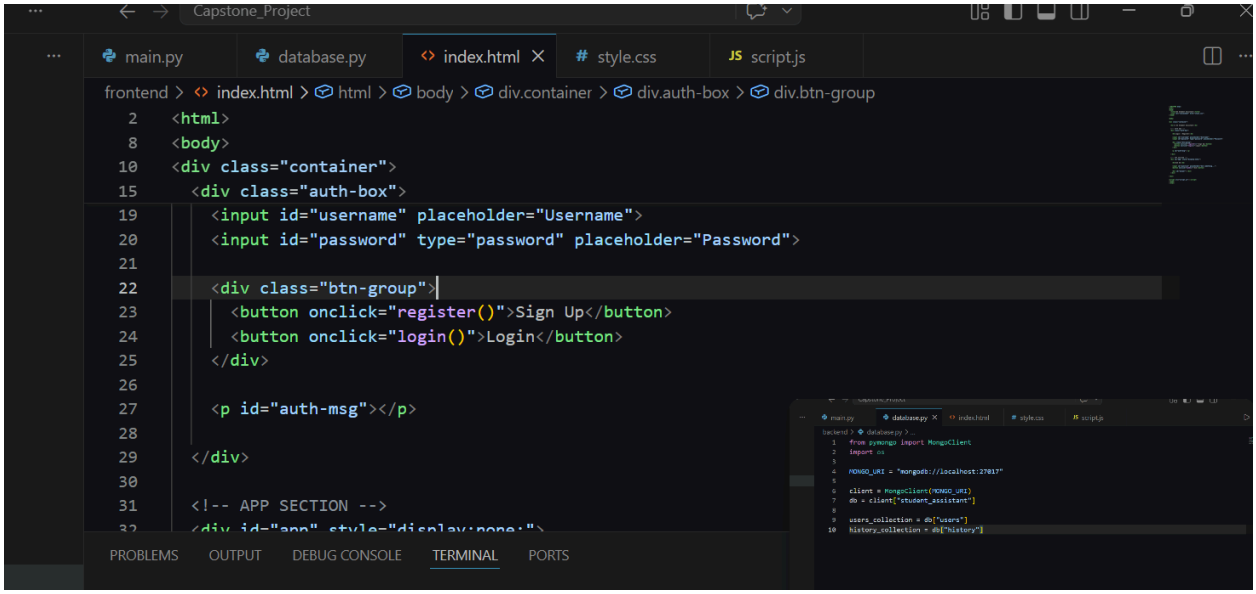
Frontend Features:

- Login/Register pages
- Dashboard for interaction
- Simple UI for questions and writing input

Code Snippet







Results

- Connected to the backend
- Successfully enter users account
- But it not complete yet
- Stores interaction history in database

Users can interact with the system in real-time and receive instant academic assistance.

Challenges

- Setting up full-stack communication
- Handling authentication securely
- Managing frontend-backend integration
- Ensuring consistent AI responses

Future Improvements

- Voice-based assistant integration
- Multilingual support
- Mobile application version
- Advanced analytics dashboard
- Personalized learning recommendations
- Progress on the current project